

Physics Assignment 5: Quiz 2 (Transformations)

My responses · 20 MCQs · Computer-graphics transformations

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1. What is the main purpose of transformations in computer graphics?

✓ To scale, translate, rotate, reflect and shear shapes and objects

- To store images
- To compress videos
- To create databases

2. Which transformation moves an object from one position to another without changing its shape?

✓ Translation

- Scaling
- Reflection
- Shearing

3. In a translation, what happens to the coordinates of an object?

✓ They are shifted by specified amounts

- They are always doubled
- They are converted into angles
- They are changed into three coordinates

4. What is the effect of increasing the x-coordinate during translation?

✓ The object moves horizontally to the right

- The object moves vertically upward
- The object rotates
- The object becomes smaller

5. What is the effect of increasing the y-coordinate during translation?

✓ The object moves vertically upward

- The object moves horizontally to the right
- The object rotates
- The object is reflected

6. Which transformation changes the size of an object?

✓ Scaling

- Translation
- Reflection
- Rotation

7. Scaling in the given 2D transformation example is performed relative to which point?

✓ The origin

- The center of the screen
- The top-left corner
- The midpoint of the object

8. What happens to a point located at the origin during scaling relative to the origin?

✓ It does not change its position

- It always moves right
- It always moves up
- It is reflected

9. Which transformation reverses the sign of the x-coordinate while keeping the y-coordinate unchanged?

✓ **Reflection about the y-axis**

- Reflection about the x-axis
- Translation
- Scaling

10. Which coordinate remains unchanged when reflecting a point about the y-axis?

✓ **The y-coordinate**

- The x-coordinate
- Both coordinates
- Neither coordinate

11. What is the basic idea of reflection in computer graphics?

✓ **Creating a mirror-like image of a shape**

- Changing only the size of a shape
- Moving a shape without changing orientation
- Converting a shape into a point

12. Homogeneous coordinates are used to represent a point in a plane using?

✓ **Three coordinates instead of two**

- One coordinate instead of two
- Two coordinates only
- Four coordinates only

13. Which of the following is a possible homogeneous representation of the point (3, 4)?

- (6, 8, 2)

✓ **(3, 4, 1)**

- (6, 4, 2)
- (3, 8, 2)

14. Why can more than one homogeneous coordinate set represent the same point?

✓ **The coordinates can be scaled proportionally**

- The point changes its location each time
- Only the first coordinate matters
- Only the second coordinate matters

15. Which mathematician is mentioned in connection with homogeneous coordinates?

✓ **August Möbius**

- Isaac Newton
- Albert Einstein
- James Clerk Maxwell

16. What is the main advantage of using homogeneous coordinates in transformations?

✓ **They provide a convenient representation for geometric transformations**

- They eliminate the need for coordinates
- They are used only for storing images
- They prevent all shape changes

17. Which transformation changes the orientation of an object by turning it through an angle?

✓ **Rotation**

- Translation
- Scaling
- Reflection

18. In 3D transformations, roll is defined as rotation about which axis?

✓ **The z-axis**

- The x-axis
- The y-axis
- No axis

19. In 3D transformations, pitch is defined as rotation about which axis?

✓ **The x-axis**

- The y-axis
- The z-axis
- No axis

20. In 3D transformations, yaw is defined as rotation about which axis?

✓ **The y-axis**

- The x-axis
- The z-axis
- No axis

Answer summary: 1-A, 2-A, 3-A, 4-A, 5-A, 6-A, 7-A, 8-A, 9-A, 10-A, 11-A, 12-A, 13-B, 14-A, 15-A, 16-A, 17-A, 18-A, 19-A, 20-A (letters follow the option order shown above).